Gauri Jain

Personal Info

Professional Interests: AI for Impact, machine learning, optimization, immigration, public health, international development
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Education

Harvard University 2022 August - 2024 December
Masters Computer Science

- Advised by Elisabeth Paulson, Milind Tambe, Berkman Klein Scholar, Mittal Institute Graduate Student Associate

Cornell University 2016 August - 2019 December
B.S. Computer Science, GPA: 3.764

- Magna Cum Laude, Tau Beta Pi member, Dean's List all Semesters

Research

1) **Gauri Jain**, Dominik Rothenhäusler, Kirk Bansak, Elisabeth Paulson. CTRL Your Shift: Clustered Transfer Residual Learning for Many Small Datasets.

- Working paper - Arxived

2) **Gauri Jain**, Pradeep Varakantham, Haifeng Xu, Aparna Taneja, Prashant Doshi, Milind Tambe. IRL Making a Difference "IRL": Inverse Reinforcement Learning for Restless Multi-Armed Bandits with Applications in Maternal and Child Health.

- PRICAI - 2024 ***Runner-up Best Paper**

3) Sean R. Sinclair, **Gauri Jain**, Siddhartha Banerjee, Christina Lee Yu. Sequential Fair Allocation: Achieving the Optimal Envy-Efficiency Trade-off Curve.

- Informs 2022 - Operations Research

4) Sean R. Sinclair, Tianyu Wang, **Gauri Jain**, Siddhartha Banerjee, Christina Lee Yu. Adaptive Discretization for Model-Based Reinforcement Learning.

- NeurIPS 2020

Work Experience

World Bank Group 2025 July - present
Consultant - Fragility, Conflict, and Violence (FCV) Analytics Group

- Directed research on the application of ML techniques to conflict monitoring, encompassing the development of escalation forecasting models and unsupervised methods for classifying and grouping conflict trajectory patterns.
- Lead development of LLM-based Risk Briefing Creation tool combining up to date FCV news with World Bank internal documents to provide operational insights

- Improve data engineering practices for LLM use in FCV group through database creation of internal documents, Text to SQL procedures, BERT models, and causal mapping data structures
- Support efforts in portfolio mapping and geo-coding, image route detection, survey response summarization for on demand management asks on mission

Flashvote 2026 Jan - present

AI Engineer

- Collaborate directly with Kevin Lyons (CEO) to build a town hall-style data platform where city government officials can see relevant data about their city when making important legislative decisions.
- Combine over 8 public datasets/data sources (ACS, HUD, NSUMHSS, FBI, CIUS, Datacommons MCP server etc.) to create useful data insights corresponding directly to users' plan language query tailored to their location (across 50 states and Puerto Rico)

Harvard Research Labs 2022 July - 2025 July

AI Researcher

- Work closely with Professors – Elisabeth Paulson, Milind Tambe (Harvard), Dominik Rothenhausler, Jens Hainmuller (Stanford), Kirk Bansak (Berkeley), Pradeep Varakantham (Singapore Management University) on AI research for impact research questions
- Collaborate with Armman and Geomatch nonprofits to develop Machine Learning research projects to directly improve their operations
- Lead teams of academics in multi-year research projects and write AI conference papers with publicly available codebases to support real-world deployment (available on website)

Armman 2023 May - 2023 August

Intern

- Work with 4 public health practitioners from Armman (nonprofit in Mumbai) and 2 researchers from Google-AI to improve maternal child health program's telehealth services for 25,000 women in Mumbai
- Develop Inverse Reinforcement Learning based algorithmic update to take health expert feedback and learn new rewards for optimal calling strategy (see paper)

Facebook 2020 May. - 2022 Aug.

Software Engineer - Marketplace Discovery + Integrity Experiences: Inform Team

- *Integrity*: Use Hack to build back-end platform for 90 election and vaccination labels seen by 3.5 billion Facebook Users
- *Marketplace*: Worked on 8 member team with software engineers, machine learning engineers, data scientists, and designers to build a new recommendation system on marketplace
- *Marketplace*: Designed, tested, and deployed experiments to 20 million Facebook Marketplace users testing different ranking backends and implemented front-end designs for better user experience

Reinforcement Learning Lab 2019 Oct. - 2020 June

Research Assistant

- Received Engaged Cornell grant to work with Operations Research Professors Siddhartha Banerjee and Christina Lee Yu as well as Ph.D. mentor Sean Sinclair.
- Collaborated with Food Bank for the Southern Tier of New York to design fair algorithm for food bank distribution in online settings using variations of Nash Welfare function. Implemented and ran experiments for two expectation based algorithms that produced less waste and were more fair than food banks current distribution
- Developed for separate project discretizing continuous state-action spaces in tree structures to efficiently store Q and Value functions. Implemented algorithm in 2 dimensional space, and adapted to ambulance placement and pendulum balancing problems.

Cornell Prison Education Program 2020 Jan. - 2020 June
Technical Consultant

- Developed Ruby on Rails application that would help coordinators keep track of over 200 students in prisons in terms of facility movement, release dates, and recidivism
- Wrote proposal detailing potential computer lab setups at the prison and gave important advice about what server and client hardware to use.
- Improved organization's website and automated future TA Application processes using Zapier and Google Sheets

AdmitHub 2018 Nov. - 2019 Feb
Software Engineering Intern

- Built an interfacing tool using Flask, Vue, and Bootstrap to allow company employees to more easily parse through chatbot mistakes in database to give students more accurate information through school chatbots
- Worked remotely with one other intern and two company engineers to build and present web-app
- Incorporated a more intuitive search function in existing mistake-finding scripts

Projects

Aliados Immigration Clinic Assistant Tool 2024 Oct. - 2026 Jan.
ML Researcher/ Developer

- Work with University of Miami Immigration Clinic to develop a compassionate interview intake chatbot with aim of reducing time burden on clinical employees and law students
- Built a full-stack chatbot application to be tested by 30 law students in 2025

The Prosocial Ranking Challenge 2024 July. - 2025 February
ML Researcher/ Developer

- Work with a team of 8 Phd students/Postdocs to develop a new ranking algorithm for Facebook, X, and Reddit with the goal of reducing polarization
- Built chrome extension that re-ranks feed by using LLM-graded insights about how surprising a post would be for political ideology
- Finalist in the competition and deployed experiment to 6000 US Social Media Users and see a 1.7% reduction in Affective polarization
- Methodology to be published

Votesee 2024 Feb. - 2024 November
Developer

- Worked with a team of 5 students from Harvard Kennedy school and Harvard Business School to build out Votesee platform which makes legislators' voting history accessible to voters
- Advised team on best practices in working with LLM API's and developed data pipeline with US Congress github to provide up-to-date voting information

Cornell Hack4Impact 2020 Jan. - 2020 June
Developer

- Worked with team of 5 other students to design and build better bus prediction for Ithaca bus system
- Used Google Transit API to access training data and designed data matrix that could consolidate data from bus routes of different sizes

- Used Random Forest algorithm to classify and adjust the current predictions for the service with a 92.5% accuracy

Interaction Lab - University of Southern California 2018 June - 2018 August
Research Assistant

- Received NSF Summer Undergraduate Experiences Grant to work with Professor Maja Mataric and 2 Ph.D. mentors
- Used ROS and Python to program a socially-assistive-robot that encourages habit-formation in elderly adults to exercise every day.
- Worked on software involved in deployment including webcam camera script and ROS action servers and clients

Teaching And Volunteer Experience

Berkman Klein Student Leaders in AI 2025 Dec - 2025 June
Leadership team

- Work with 3 other Harvard students on interdisciplinary leadership team to create a community of 45 scholars across 12 Harvard schools for semester long speaker series and discussion
- Organize speaker sessions, and moderate panels with researchers and practitioners from the US government, Harvard/MIT, Industry professionals, etc.

Harvard AI for Impact, Fundamental of CS courses 2024 Jan - 2024 June
Teaching Fellow

- Work closely with Professors Milind Tambe and Harry Lewis to design coursework
- Support course administration, grading, and designing HW/Readings/speaker series in courses
- Advise Harvard undergrad, masters, and Phd students on final projects/AI for Impact papers

Code In Place - Stanford University 2020 March - 2021 June
Section Leader

- Part of a teaching team for Code in Place, offered by Stanford during COVID-19 pandemic, with 10,000 global students and 900 volunteer teachers participating from around the world.
- Prepared and taught a weekly discussion section of 10-12 students to supplement professors' lectures in a 5-week introductory online Python programming course based on material from the first half of Stanford's introductory programming course, CS106A

Breakthrough Middle School Summer Program 2020 June - 2020 August
Teaching Fellow

- Develop lesson plans for 6 week Computer Science, Algebra A, and Algebra B courses for middle schoolers at virtual summer program
- Taught courses with 8-15 students over Zoom, wrote/graded HW, planned online scavenger hunts, jeopardy games, and other engaging activities to overcome online program
- In CS course, had students build their own personalized games on Scratch and taught them about one emerging technology per day: Boston Dynamics robots, graphics in animated movies, reinforcement learning in gaming competitions, stable matching algorithms, etc.

Cornell Prison Education Program 2019 Aug. - 2020 March
Physics/Computer Lab TA

- Worked as TA for 12 students at Auburn Correctional Facility at upstate New York

- Created printouts of HW answer keys for all students with detailed explanations to accommodate for low resources in the prison
- Ran review sessions and help students in an office hours format.

Women in Computing at Cornell 2018 August- 2020 May
Faculty and Alumni Director

- Organize weekly events connecting faculty and alumni in computing with students
- Plan Alumni networking event, Faculty Lightning Talk, and lunch with faculty members and underclassmen

Workshop Papers + Posters

1) **Gauri Jain***, Pradeep Varakantham, Haifeng Xu, Aparna Taneja, Milind Tambe. Inverse Reinforcement Learning for Restless Multi-Armed Bandits with Application to Maternal and Child Health.

- Accepted as paper at IJCAI Workshop on Bridge-AI: from Climate Change to Health Equity (2023)

2) Ziwei Gu*, **Gauri Jain***, Hongwen Song*, Isak Diaz, Margaux Masson-Forsythe, Jorge Valdes. BiomeAzuero2022: A Fine-Grained Dataset and Baselines for Tree Species Classification with Ground Images.

- Accepted as poster at AAAI Workshop on AI for Social Good (2023)

3) Sean R. Sinclair*, **Gauri Jain**, Siddhartha Banerjee, Christina Lee Yu. Sequential Fair Allocation of Limited Resources under Stochastic Demands.

- Accepted at Harvard CRCS Workshop on AI for Social Good (2020)
- Accepted at Mechanism Design for Social Good Workshop (2020)

4) Sean R. Sinclair*, Tianyu Wang*, **Gauri Jain**, Siddhartha Banerjee, Christina Lee Yu. Adaptive Discretization for Model-Based Reinforcement Learning.

- Accepted at ICML Workshop on Theoretical Foundations of Reinforcement Learning (2020)

5) **Gauri Jain***, Audrow Nash, Katelyn Swift-Pong, Maja Mataric. Toward a Pose Detection Robot System for Physical Rehabilitation (Poster).

- Presented at University of Southern California Summer NSF Research Symposium (2018)

6) **Gauri Jain***, Alexis Crawshaw, Nathan Gugel. Assessing the Vibrational Effectiveness and Design Constraints of Sensitivity-Modified Transducers on a Vibrotactile Music Suit.

- Submitted at the 2015 Research Mentorship Program at University of California Santa Barbara. Accompanied by a 10 minute presentation to audience of 100 faculty, researchers, and parents (2015)

Skills and Coursework

Coursework AI for Social Impact, Inverse Reinforcement Learning for Healthcare Applications,Graduate Algorithms, Machine Learning, Language and Information,Computer Vision, Natural Language Processing, Object Oriented Programming and Data Structures, Functional Programming, Operating Systems, Discrete Structures, Probability and Inference, Linear Algebra

Programming Python, Numpy, Jupyter, Matplotlib, Pandas, Java, C/C++, OCaml, Javascript/CSS/HTML, SQL, React, Hack, Ruby on Rails

Languages English, Hindi

About Me

I enjoy playing music (piano, guitar, singing) and also love exercising, playing sports, and being outdoors! I also adore pitbulls and feel most at peace with a cup of chai in hand.